Multi-Tier AWS Storage Deployment for Data Engineering Workloads

[AWS Data Engineering Project - 1]

1. Project Overview

The goal of this project is to design and implement a **multi-tier AWS storage architecture** for a data engineering workflow. The solution will leverage:

- **EBS (Elastic Block Store)** for transactional database storage.
- **EFS (Elastic File System)** for shared logs and application data.
- **S3 (Simple Storage Service)** for backups and archival.
- Additional AWS services like **EC2, VPC, IAM, and Autoscaling** for a production-grade deployment.

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2. Project Scope

Core Components

1. **Virtual Private Cloud (VPC) Setup**
    - Public and private subnets across multiple AZs.
    - NAT Gateway for private subnet internet access.
    - Security groups and NACLs for network security.
2. **Storage Tier Implementation**
    - **EBS**: Attached to EC2 instances running a relational database (MySQL/PostgreSQL).
    - **EFS**: Mounted on multiple EC2 instances for shared log storage.
    - **S3**: Used for automated backups of EBS snapshots and EFS data.
3. **Compute & Automation**
    - **EC2 Auto Scaling Group (ASG)** for high availability.
    - **Application Load Balancer (ALB)** to distribute traffic.
    - **Lambda** for backup automation (optional).
4. **Security & Monitoring**
    - IAM roles with least privilege.
    - CloudWatch for logging and alerts.

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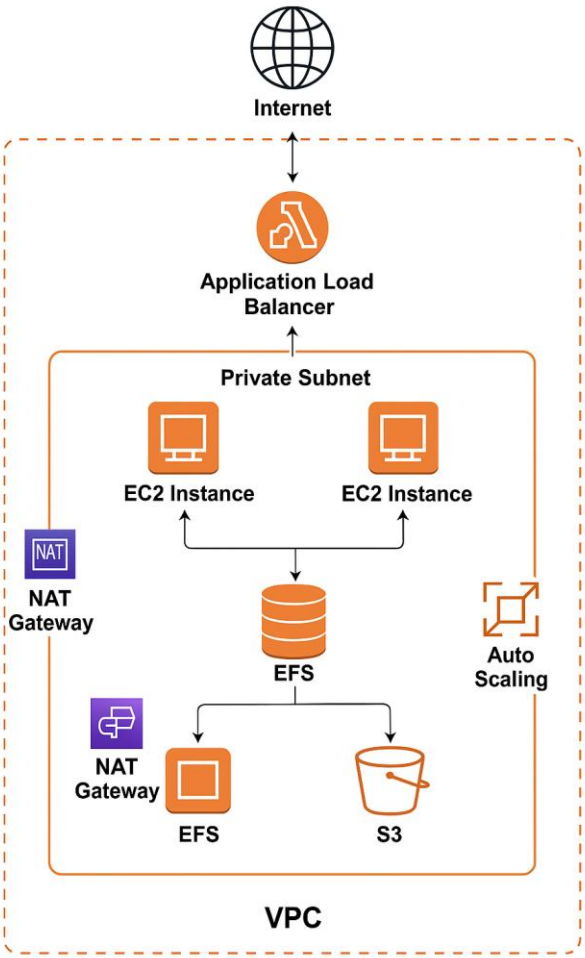
3. Deliverables

A. Architecture Diagram

Visual Representation:



Description:

- **VPC** with public/private subnets.
- **EBS** volumes for database (MySQL).
- **EFS** mounted on web/app servers.
- **S3** for backups.
- **ALB + ASG** for scalability.

B. Step-by-Step Implementation with Screenshots

1. VPC & Networking Setup

✅ **Step 1.1 – Create a Custom VPC**

- Define a CIDR block (e.g., 10.0.0.0/16).
- Enable DNS hostname and resolution.

✅ **Step 1.2 – Create Subnets**

- **Public Subnet A & B** (e.g., 10.0.1.0/24, 10.0.2.0/24)
- **Private Subnet A & B** (e.g., 10.0.3.0/24, 10.0.4.0/24)

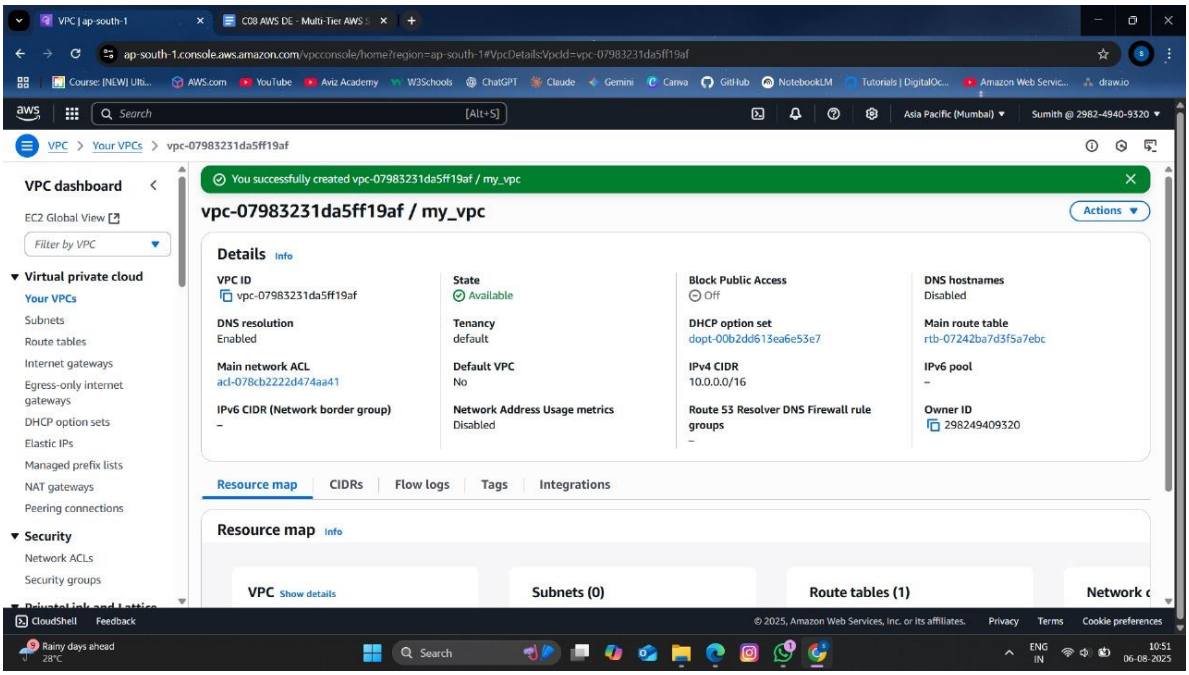
✅ **Step 1.3 – Configure Internet Gateway & NAT**

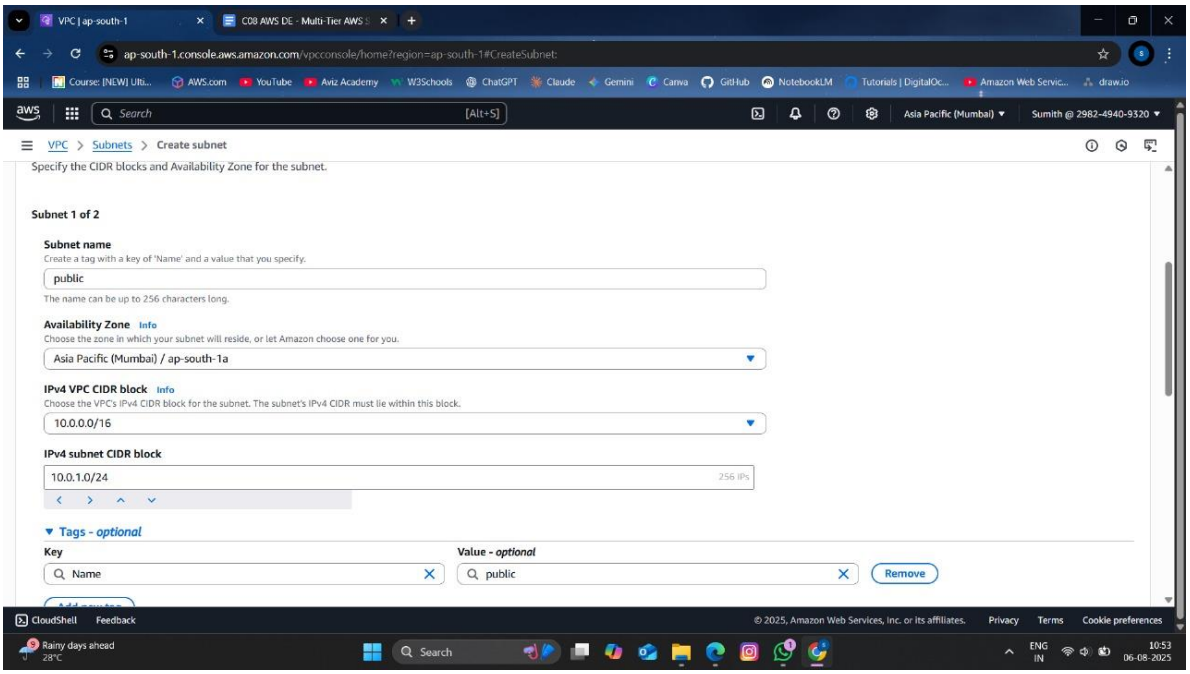
- Create and attach an **Internet Gateway** to the VPC.
- Create a **NAT Gateway** in each public subnet.
- Add route tables:
  - Public subnets route to IGW.
  - Private subnets route to NAT Gateway for internet access.

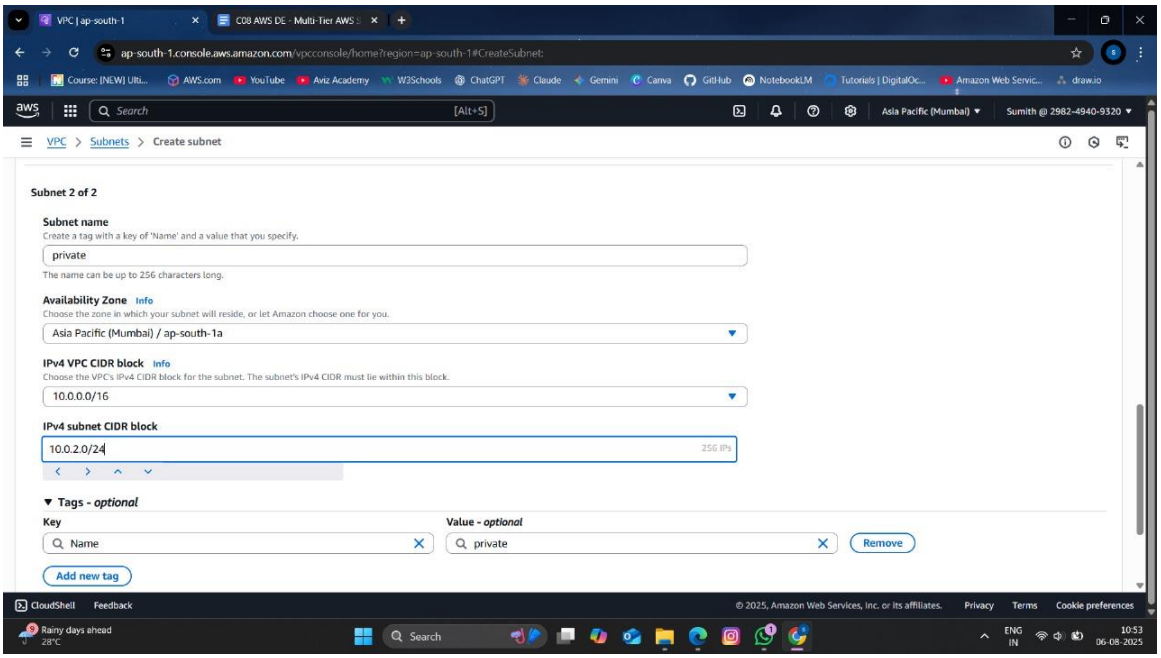
✅ **Step 1.4 – Configure Security**

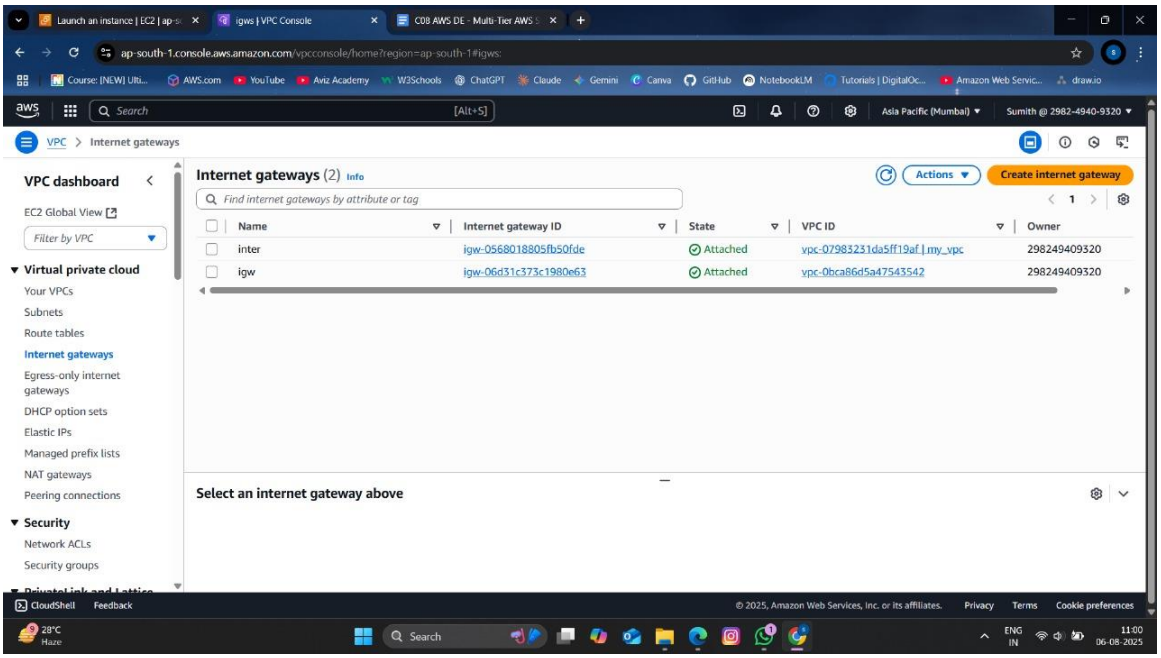
- **Security Groups**:
  - ALB: Allow inbound HTTP/HTTPS.
  - EC2: Allow from ALB, restrict SSH.
- **Network ACLs**: Add stateless layer for subnet-level filtering.

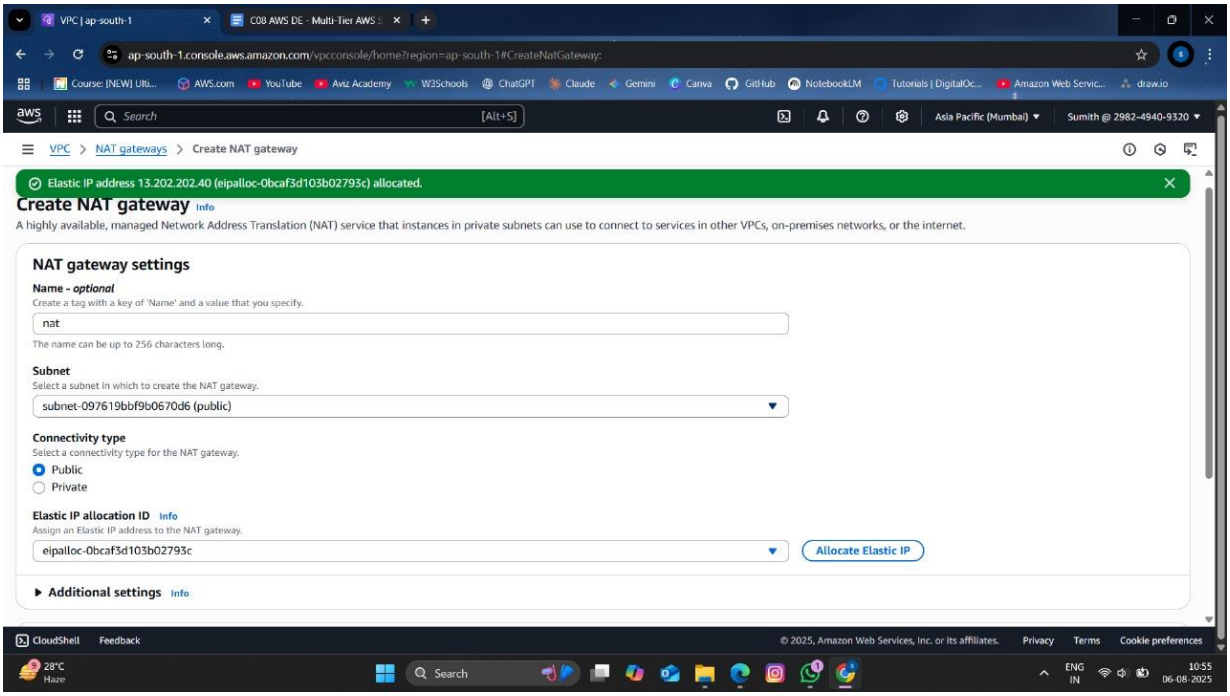
Screenshot:

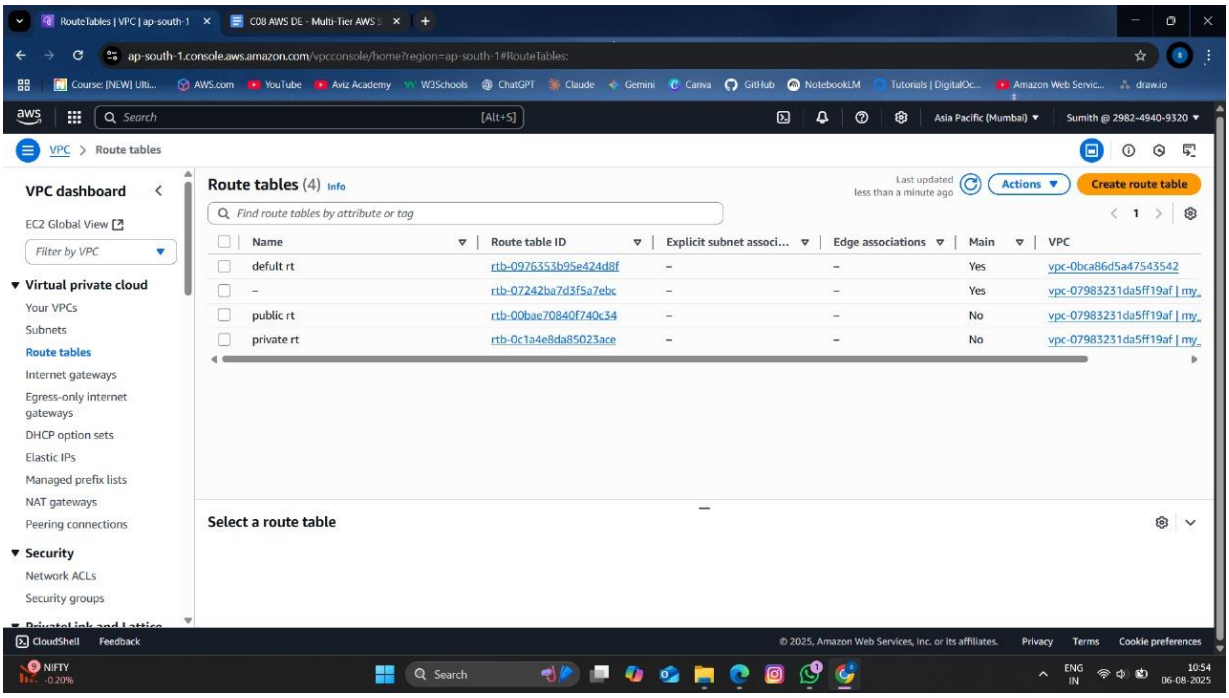



VPC | ap-south-1
C08 AWS DE - Multi-Tier AWS
ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#CreateSubnet:
Asia Pacific (Mumbai)
Sumith @ 2982-4940-9320
VPC > Subnets > Create subnet
Specify the CIDR blocks and Availability Zone for the subnet.
Subnet 1 of 2
Subnet name
Create a tag with a key of 'Name' and a value that you specify.
public
The name can be up to 256 characters long.
Availability Zone Info
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
Asia Pacific (Mumbai) / ap-south-1a
IPv4 VPC CIDR block Info
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
10.0.0.0/16
IPv4 subnet CIDR block
10.0.1.0/24
256 IPs
Tags - optional
Key
Value - optional
Name
public
Remove
CloudShell
Feedback
© 2025, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


VPC | ap-south-1
C08 AWS DE - Multi-Tier AWS
ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#CreateSubnet:
Asia Pacific (Mumbai)
Sumith @ 2982-4940-9320
VPC > Subnets > Create subnet
Subnet 2 of 2
Subnet name
Create a tag with a key of 'Name' and a value that you specify.
private
The name can be up to 256 characters long.
Availability Zone Info
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
Asia Pacific (Mumbai) / ap-south-1a
IPv4 VPC CIDR block Info
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
10.0.0.0/16
IPv4 subnet CIDR block
10.0.2.0/24
256 IPs
Tags - optional
Key
Value - optional
Name
private
Remove
Add new tag
CloudShell
Feedback
© 2025, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


Launch an instance | EC2 | ap-s
igws | VPC Console
C08 AWS DE - Multi-Tier AWS S
ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#igws:
Course: [NEW] Ulti...
AWS.com
YouTube
Aviz Academy
W3Schools
ChatGPT
Claude
Gemini
Canva
GitHub
NotebookLM
Tutorials | DigitalOc...
Amazon Web Servic...
draw.io
Search
[Alt+S]
Asia Pacific (Mumbai)
Sumith @ 2982-4940-9320
VPC > Internet gateways
VPC dashboard
EC2 Global View
Filter by VPC
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Security
Network ACLs
Security groups
Internet gateways (2) Info
Actions
Create internet gateway
Find internet gateways by attribute or tag
Name
Internet gateway ID
State
VPC ID
Owner
inter
igw-0568018805fb50fde
Attached
vpc-07983231da5ff19af | my_vpc
298249409320
igw
igw-06d31c373c1980e63
Attached
vpc-0bca86d5a47543542
298249409320
Select an internet gateway above
CloudShell
Feedback
© 2025, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
28°C
Haze
Search
ENG
IN
11:00
06-08-2025


VPC | ap-south-1
C08 AWS DE - Multi-Tier AWS S
ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#CreateNatGateway:
Course: [NEW] Ulti...
AWS.com
YouTube
Aviz Academy
W3Schools
ChatGPT
Claude
Gemini
Canva
GitHub
NotebookLM
Tutorials | DigitalOc...
Amazon Web Servic...
draw.io
Search
[Alt+S]
Asia Pacific (Mumbai)
Sumith @ 2982-4940-9320
VPC > NAT gateways > Create NAT gateway
Elastic IP address 13.202.202.40 (eipalloc-0bcaf3d103b02793c) allocated.
Create NAT gateway Info
A highly available, managed Network Address Translation (NAT) service that instances in private subnets can use to connect to services in other VPCs, on-premises networks, or the internet.
NAT gateway settings
Name - optional
Create a tag with a key of 'Name' and a value that you specify.
nat
The name can be up to 256 characters long.
Subnet
Select a subnet in which to create the NAT gateway.
subnet-097619bbf9b0670d6 (public)
Connectivity type
Select a connectivity type for the NAT gateway.
Public
Private
Elastic IP allocation ID Info
Assign an Elastic IP address to the NAT gateway.
eipalloc-0bcaf3d103b02793c
Allocate Elastic IP
Additional settings Info
CloudShell
Feedback
© 2025, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
28°C
Haze
Search
ENG
IN
10:55
06-08-2025



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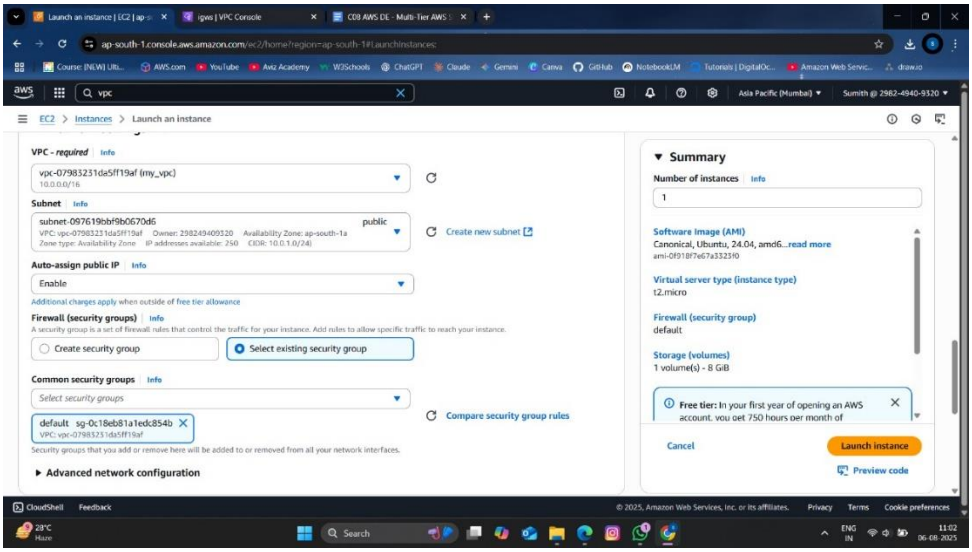
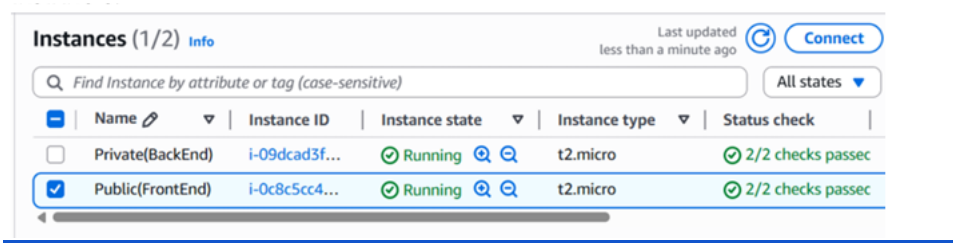
2. EC2 & EBS Deployment

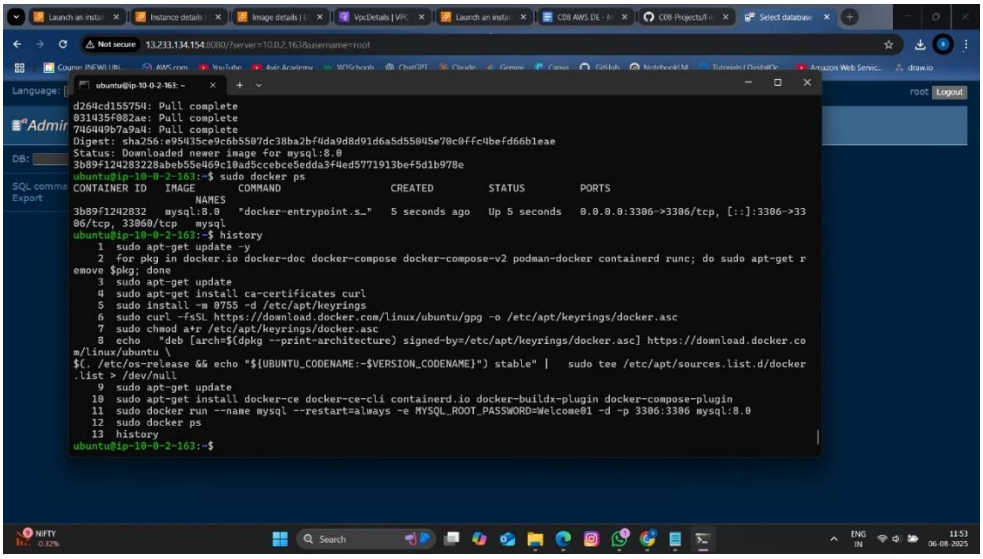
Steps:

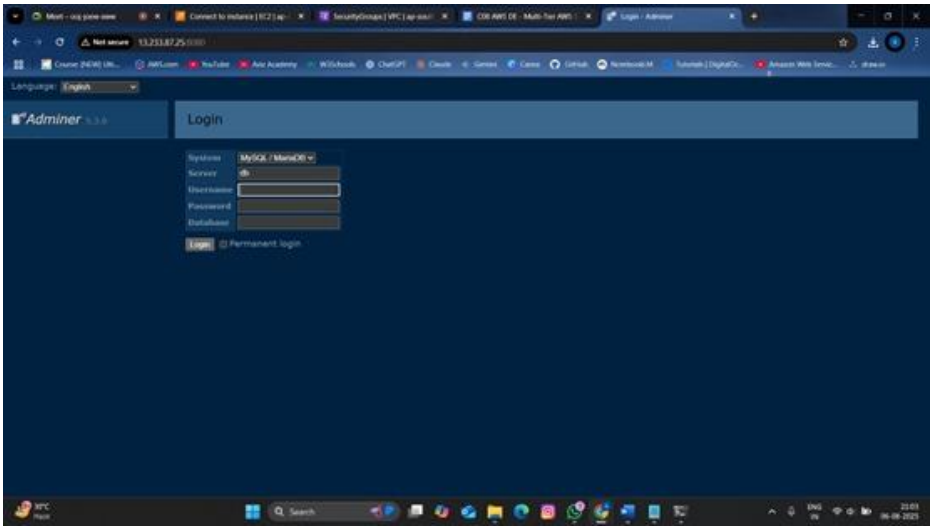
1. Launch an EC2 instance in the private subnet for MySQL.
2. Attach a **GP3 EBS volume** (50 GB) for database storage.
3. Install MySQL and verify connectivity.

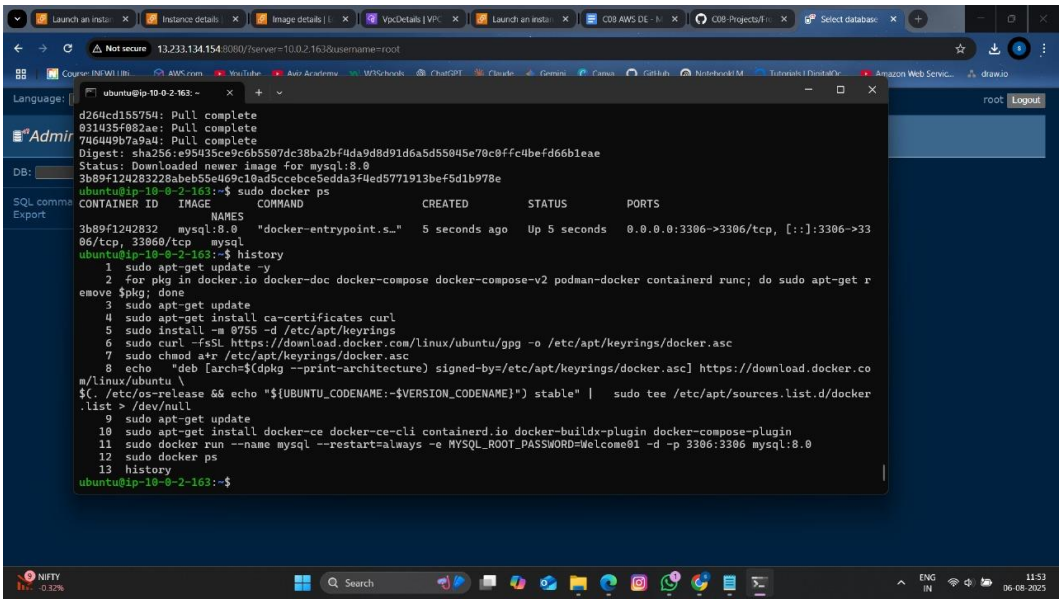
Screenshot:

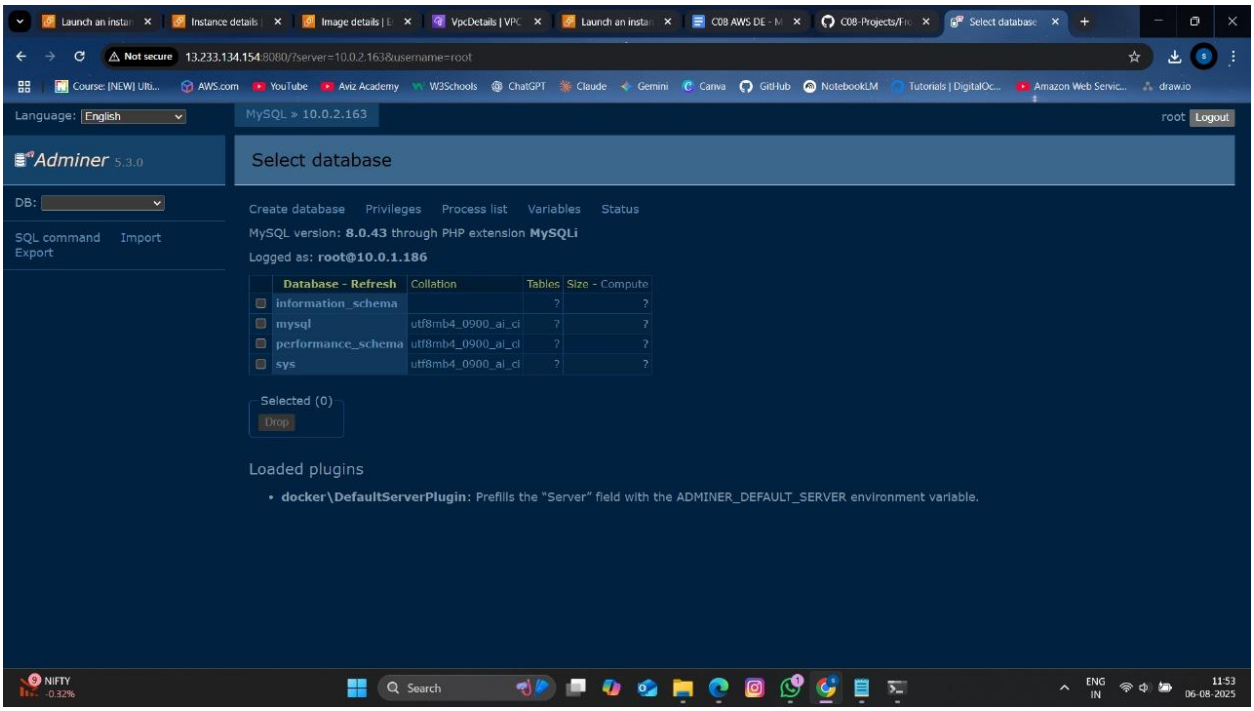

EC2 > Instances > Launch an instance
VPC - required
vpc-07983231da5ff19af (my_vpc)
Subnet
subnet-097619bbf9b0670d6
Create new subnet
Auto-assign public IP
Enable
Firewall (security groups)
Create security group
Select existing security group
Common security groups
Select security groups
default sg-0c18eb81a1edc854b
Compare security group rules
Advanced network configuration
Summary
Number of instances
1
Software Image (AMI)
Canonical, Ubuntu, 24.04, amd6...read more
Virtual server type (instance type)
t2.micro
Firewall (security group)
default
Storage (volumes)
1 volume(s) - 8 GiB
Free tier: In your first year of opening an AWS account, you get 750 hours per month of
Cancel
Launch instance
Preview code


Instances (1/2) Info
Last updated less than a minute ago
Connect
Find Instance by attribute or tag (case-sensitive)
All states
Name
Instance ID
Instance state
Instance type
Status check
Private(BackEnd)
i-09dcad3f...
Running
t2.micro
2/2 checks passed
Public(FrontEnd)
i-0c8c5cc4...
Running
t2.micro
2/2 checks passed


ubuntu@ip-10-0-2-163: ~
d264cd155754: Pull complete
031435f082ae: Pull complete
746449b7a9a4: Pull complete
Digest: sha256:e95435ce9c6b5507dc38ba2bf4da9d8d91d6a5d55045e70c0ffc4befd66b1eae
Status: Downloaded newer image for mysql:8.0
3b89f1242832abeb55e469c10ad5ccebce5edda3f4ed5771913bef5d1b978e
ubuntu@ip-10-0-2-163:~$ sudo docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
3b89f1242832 mysql:8.0 "docker-entrypoint.s..." 5 seconds ago Up 5 seconds 0.0.0.0:3306->3306/tcp, [::]:3306->3306/tcp, 33060/tcp mysql
ubuntu@ip-10-0-2-163:~$ history
1 sudo apt-get update -y
2 for pkg in docker.io docker-doc docker-compose docker-compose-v2 podman-docker containerd runc; do sudo apt-get remove $pkg; done
3 sudo apt-get update
4 sudo apt-get install ca-certificates curl
5 sudo install -m 0755 -d /etc/apt/keyrings
6 sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg -o /etc/apt/keyrings/docker.asc
7 sudo chmod a+r /etc/apt/keyrings/docker.asc
8 echo "deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.asc] https://download.docker.com/linux/ubuntu \
$(. /etc/os-release && echo "${UBUNTU_CODENAME:-$VERSION_CODENAME}") stable" | sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
9 sudo apt-get update
10 sudo apt-get install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
11 sudo docker run --name mysql --restart=always -e MYSQL_ROOT_PASSWORD=Welcome01 -d -p 3306:3306 mysql:8.0
12 sudo docker ps
13 history
ubuntu@ip-10-0-2-163:~$


Adminer
Login
Language: English
System
MySQL / MariaDB
Server
db
Username
Password
Database
Login
Permanent login


Not secure
13.233.134.154:8080/?server=10.0.2.163&username=root
ubuntu@ip-10-0-2-163: ~
d264cd155754: Pull complete
031435f082ae: Pull complete
746449b7a9a4: Pull complete
Digest: sha256:e95435ce9c6b5507dc38ba2bf4da9d8d91d6a5d55045e70c0ffc4befd66b1eae
Status: Downloaded newer image for mysql:8.0
3b89f124283228abeb55e469c10ad5ccebce5edda3f4ed5771913bef5d1b978e
ubuntu@ip-10-0-2-163:~$ sudo docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS
NAMES
3b89f1242832 mysql:8.0 "docker-entrypoint.s…" 5 seconds ago Up 5 seconds 0.0.0.0:3306->3306/tcp, [::]:3306->3306/tcp, 33060/tcp mysql
ubuntu@ip-10-0-2-163:~$ history
1 sudo apt-get update -y
2 for pkg in docker.io docker-doc docker-compose docker-compose-v2 podman-docker containerd runc; do sudo apt-get remove $pkg; done
3 sudo apt-get update
4 sudo apt-get install ca-certificates curl
5 sudo install -m 0755 -d /etc/apt/keyrings
6 sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg -o /etc/apt/keyrings/docker.asc
7 sudo chmod a+r /etc/apt/keyrings/docker.asc
8 echo "deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.asc] https://download.docker.com/linux/ubuntu \
$(. /etc/os-release && echo "${UBUNTU_CODENAME:-$VERSION_CODENAME}") stable" | sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
9 sudo apt-get update
10 sudo apt-get install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
11 sudo docker run --name mysql --restart=always -e MYSQL_ROOT_PASSWORD=Welcome01 -d -p 3306:3306 mysql:8.0
12 sudo docker ps
13 history
ubuntu@ip-10-0-2-163:~$
root
Logout

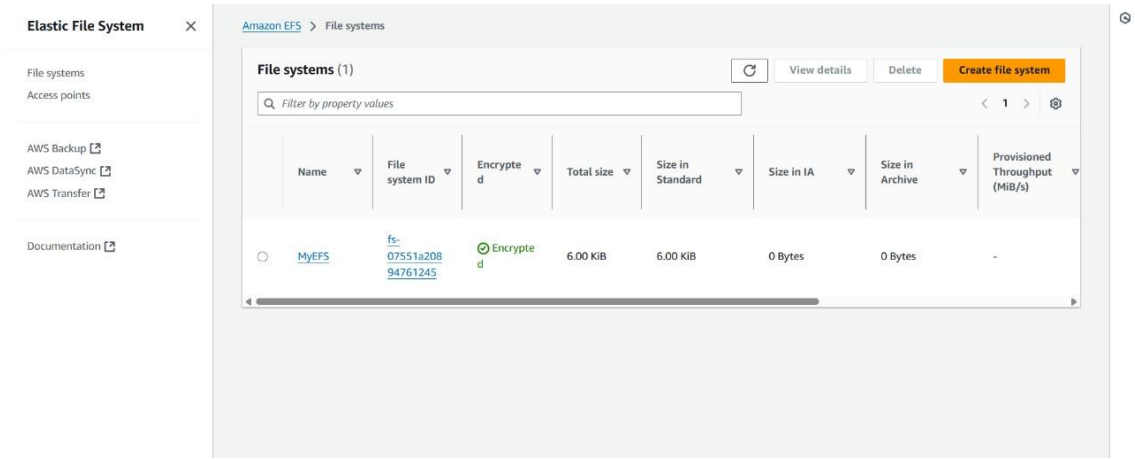


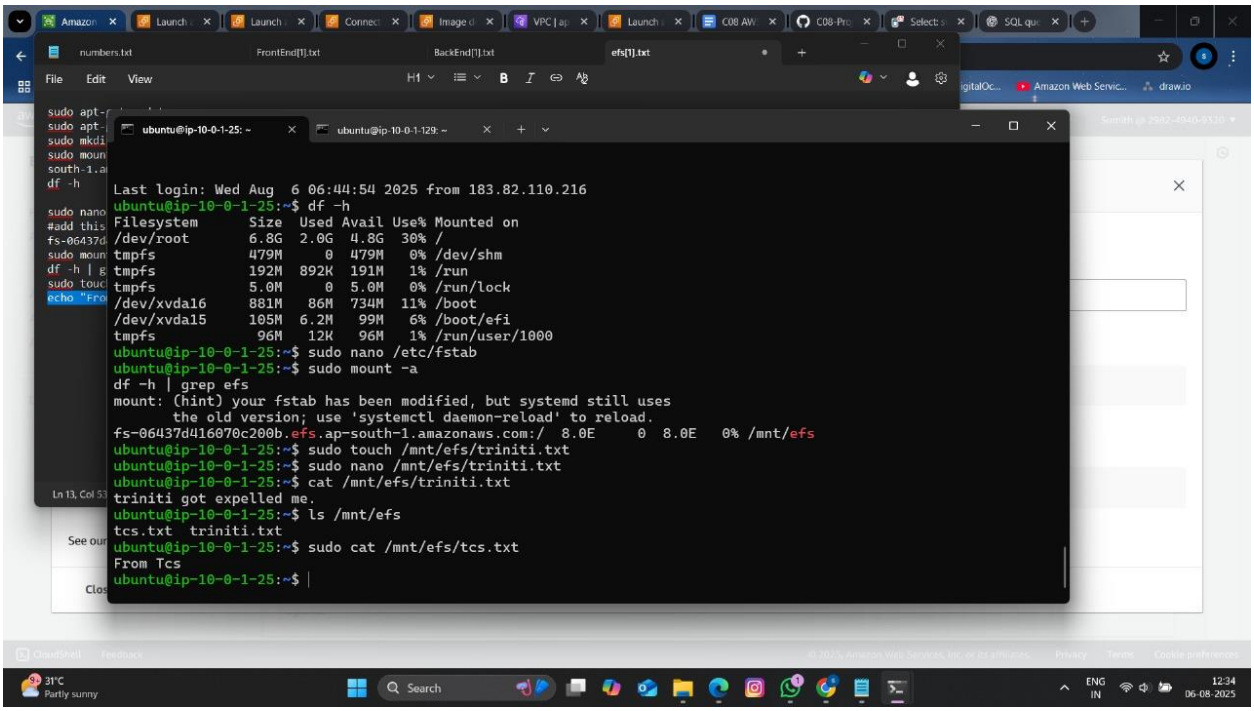
3. EFS Setup for Shared Storage

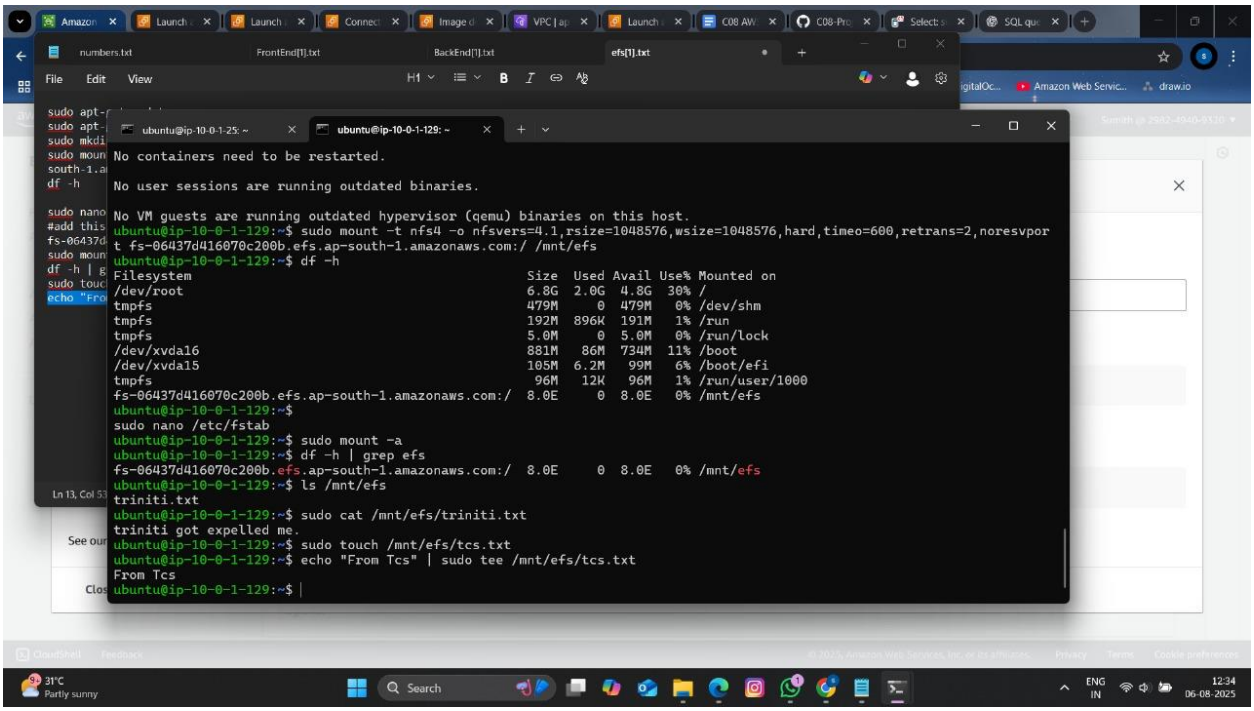
Steps:

1. Create an **EFS file system** in the same VPC.
2. Mount EFS on multiple EC2 instances (e.g., web servers).
3. Test file sharing between instances.

Screenshot:


Elastic File System
File systems
Access points
AWS Backup
AWS DataSync
AWS Transfer
Documentation
Amazon EFS > File systems
File systems (1)
View details
Delete
Create file system
Filter by property values
Name
File system ID
Encrypted
Total size
Size in Standard
Size in IA
Size in Archive
Provisioned Throughput (MiB/s)
MyEFS
fs-07551a20894761245
Encrypted
6.00 KiB
6.00 KiB
0 Bytes
0 Bytes
-


ubuntu@ip-10-0-1-25: ~
ubuntu@ip-10-0-1-129: ~
Last login: Wed Aug 6 06:44:54 2025 from 183.82.110.216
ubuntu@ip-10-0-1-25:~$ df -h
Filesystem Size Used Avail Use% Mounted on
/dev/root 6.8G 2.0G 4.8G 30% /
tmpfs 479M 0 479M 0% /dev/shm
tmpfs 192M 892K 191M 1% /run
tmpfs 5.0M 0 5.0M 0% /run/lock
/dev/xvda16 881M 86M 734M 11% /boot
/dev/xvda15 105M 6.2M 99M 6% /boot/efi
tmpfs 96M 12K 96M 1% /run/user/1000
ubuntu@ip-10-0-1-25:~$ sudo nano /etc/fstab
ubuntu@ip-10-0-1-25:~$ sudo mount -a
df -h | grep efs
mount: (hint) your fstab has been modified, but systemd still uses
the old version; use 'systemctl daemon-reload' to reload.
fs-06437d416070c200b.efs.ap-south-1.amazonaws.com:/ 8.0E 0 8.0E 0% /mnt/efs
ubuntu@ip-10-0-1-25:~$ sudo touch /mnt/efs/triniti.txt
ubuntu@ip-10-0-1-25:~$ sudo nano /mnt/efs/triniti.txt
ubuntu@ip-10-0-1-25:~$ cat /mnt/efs/triniti.txt
triniti got expelled me.
ubuntu@ip-10-0-1-25:~$ ls /mnt/efs
tcs.txt triniti.txt
ubuntu@ip-10-0-1-25:~$ sudo cat /mnt/efs/tcs.txt
From Tcs
ubuntu@ip-10-0-1-25:~$



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4. S3 Backup Automation

Steps:

1. Create an S3 bucket with versioning enabled.
2. Use **AWS Backup** or a Lambda function to:
    - Take daily EBS snapshots.
    - Sync EFS data to S3.

Screenshot:

https://ap-south-1.console.aws.amazon.com/s3/bucket/create?region=ap-south-1&bucketType=general

Amazon S3 > Buckets > Create bucket

Create bucket Info

Buckets are containers for data stored in S3.

General configuration

AWS Region

Asia Pacific (Mumbai) ap-south-1

Bucket type Info

- General purpose — Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.
- Directory — Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.

Bucket name Info

sumith-0

Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). Learn More

Copy settings from existing bucket - *optional*

Only the bucket settings in the following configuration are copied.

Choose bucket

Format: s3://bucket/prefix

Object Ownership Info

Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.

https://ap-south-1.console.aws.amazon.com/s3/buckets/sumith-0?region=ap-south-1&bucketType=general&tab=properties

Amazon S3 > Buckets > sumith-0

sumith-0 Info

Objects | Properties | Permissions | Metrics | Management | Access Points

Bucket overview

AWS Region	Amazon Resource Name (ARN)	Creation date
Asia Pacific (Mumbai) ap-south-1	arn:aws:s3:::sumith-0	August 6, 2025, 21:59:13 (UTC+05:30)

Bucket Versioning

Edit

Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can easily recover from both unintended user actions and application failures. Learn more

Bucket Versioning

Disabled

Multi-factor authentication (MFA) delete

An additional layer of security that requires multi-factor authentication for changing Bucket Versioning settings and permanently deleting object versions. To modify MFA delete settings, use the AWS CLI, AWS SDK, or the Amazon S3 REST API. Learn more

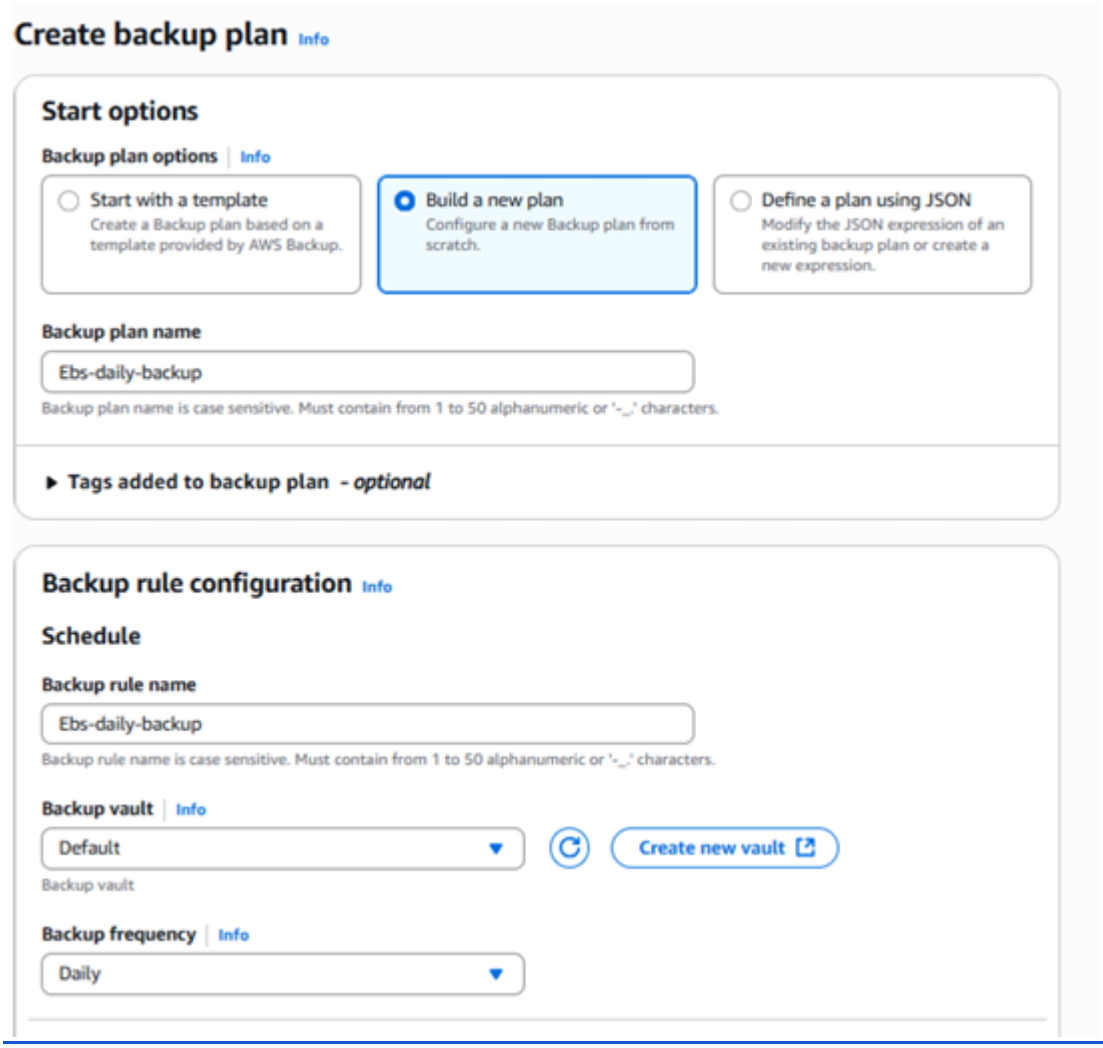
Disabled

Tags (0)

Edit

You can use bucket tags to track storage costs and organize buckets. Learn more

AWS Backup:


Create backup plan Info
Start options
Backup plan options Info
Start with a template
Create a Backup plan based on a template provided by AWS Backup.
Build a new plan
Configure a new Backup plan from scratch.
Define a plan using JSON
Modify the JSON expression of an existing backup plan or create a new expression.
Backup plan name
Ebs-daily-backup
Backup plan name is case sensitive. Must contain from 1 to 50 alphanumeric or '-_.' characters.
Tags added to backup plan - optional
Backup rule configuration Info
Schedule
Backup rule name
Ebs-daily-backup
Backup rule name is case sensitive. Must contain from 1 to 50 alphanumeric or '-_.' characters.
Backup vault Info
Default
Create new vault
Backup vault
Backup frequency Info
Daily

AWS Backup > Backup plans > Ebs-daily-backup > Assign resources

Assign resources Info

General

Resource assignment name

Daily-EBS-Backup-Plan

Resource assignment name is case sensitive. Must contain from 1 to 50 alphanumeric or '-_' characters.

IAM role | Info

AWS Backup will assume this IAM role when creating and managing recovery points on your behalf.

- ○ Default role
  If the AWS Backup default role is not present, one will be created for you with the correct permissions.
- ◉ Choose an IAM role

Role name

Role_For_EBS_Backup

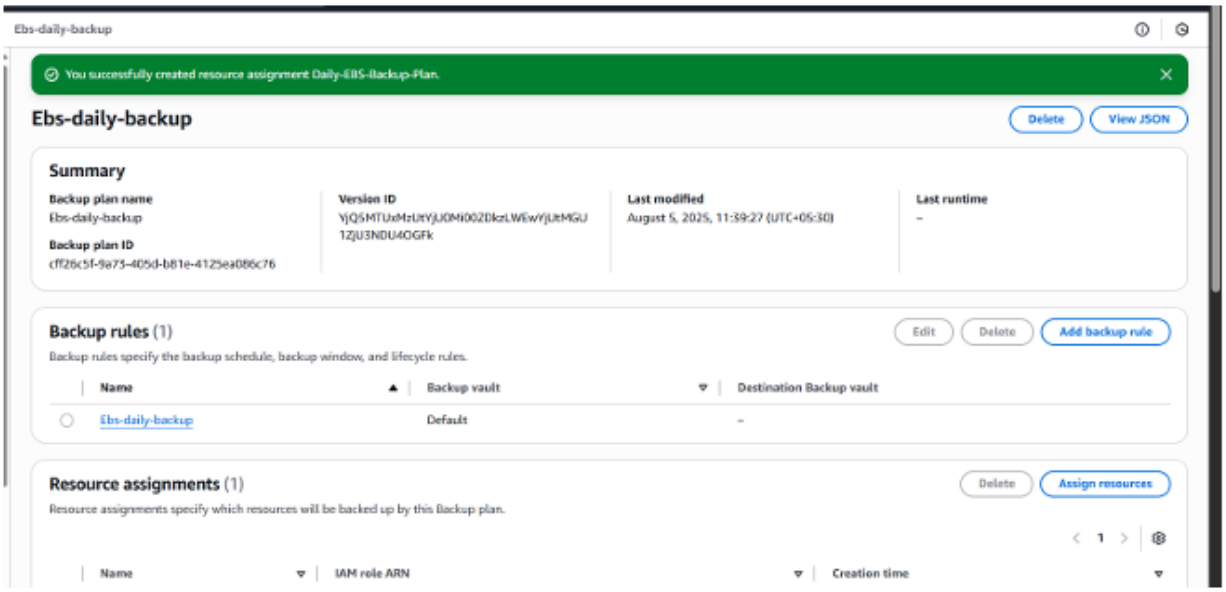
2. Select specific resource types Info

Choose specific resource types that you want to protect with this backup plan. You can also exclude specific resource IDs from the selection.

Select resource types

Resource type	Volume IDs	
EBS	*Choose resources*	Remove
	All volumes ✕	

Resource type	File system IDs	
EFS	*Choose resources*	Remove
	All file systems ✕	



Backup Plan Summary:

• Plan Name: aws/efs/automatic-backup-plan

• Backup Vault: default

• Backup Rule: automatic-backup-rule

• Resource Assignments:

o MYEFS(EFS)

o Daily Ebs BackupPlan

• IAM Roles:

o Correct service roles are assigned (Role_For_EBS_Backup)

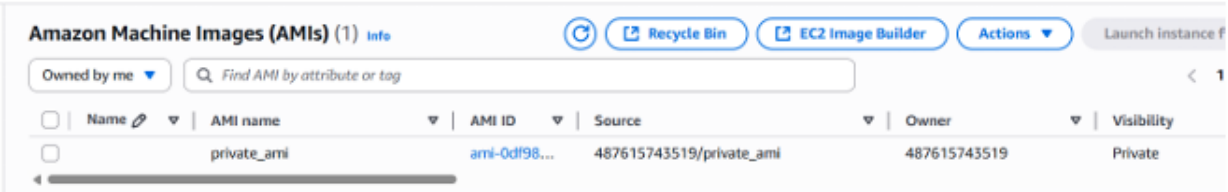
---

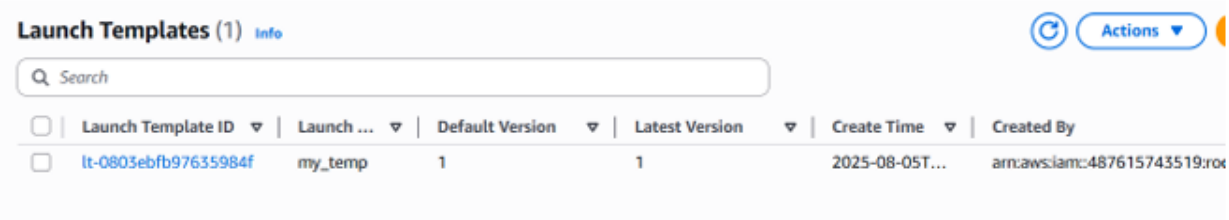
5. Autoscaling & Load Balancer

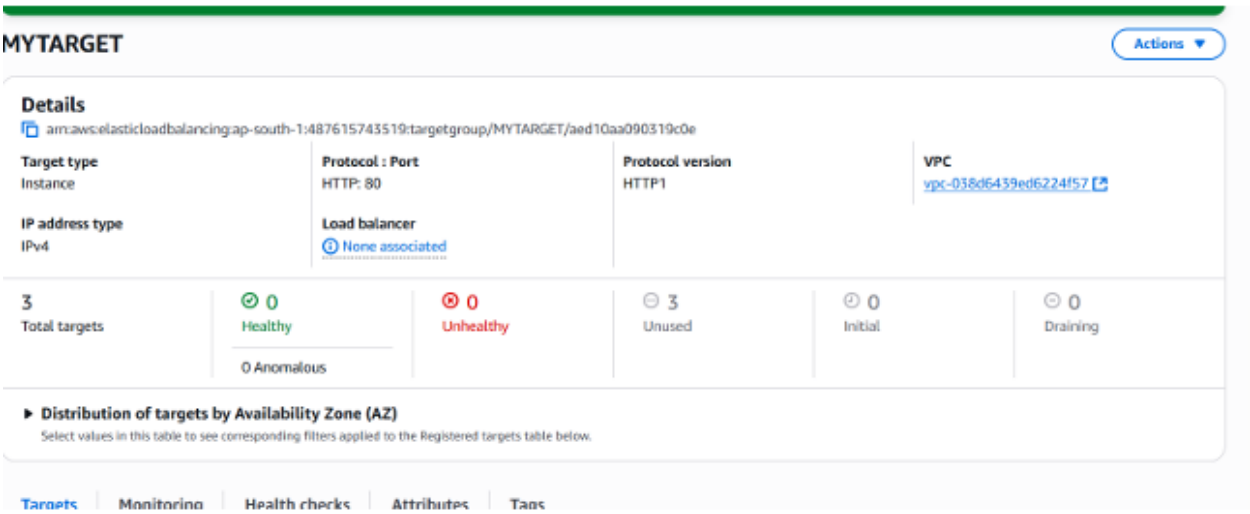
Steps:

1. Create a **Launch Template** with EFS mount instructions.
2. Configure an **ASG** with min/max capacity.
3. Set up an **ALB** to route traffic.

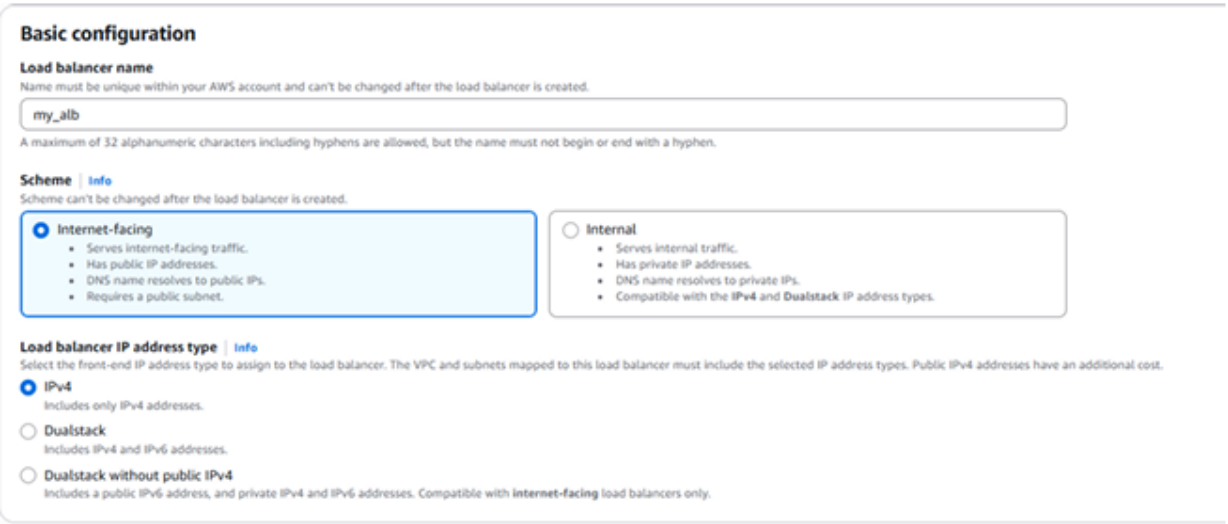
Screenshot:


Amazon Machine Images (AMIs) (1) Info
Recycle Bin
EC2 Image Builder
Actions
Owned by me
Find AMI by attribute or tag
Name
AMI name
AMI ID
Source
Owner
Visibility
private_ami
ami-0df98...
487615743519/private_ami
487615743519
Private


Launch Templates (1) Info
Actions
Search
Launch Template ID
Launch ...
Default Version
Latest Version
Create Time
Created By
lt-0803ebfb97635984f
my_temp
1
1
2025-08-05T...


MYTARGET
Actions
Details
arn:aws:elasticloadbalancing:ap-south-1:487615743519:targetgroup/MYTARGET/aed10aa090319c0e
Target type
Instance
Protocol : Port
HTTP: 80
Protocol version
HTTP1
VPC
vpc-038d6439ed6224f57
IP address type
IPv4
Load balancer
None associated
3
Total targets
0
Healthy
0 Anomalous
0
Unhealthy
3
Unused
0
Initial
0
Draining
Distribution of targets by Availability Zone (AZ)
Select values in this table to see corresponding filters applied to the Registered targets table below.
Targets
Monitoring
Health checks
Attributes
Tags

ALB:


Basic configuration
Load balancer name
Name must be unique within your AWS account and can't be changed after the load balancer is created.
my_alb
A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.
Scheme Info
Scheme can't be changed after the load balancer is created.
Internet-facing
Serves internet-facing traffic.
Has public IP addresses.
DNS name resolves to public IPs.
Requires a public subnet.
Internal
Serves internal traffic.
Has private IP addresses.
DNS name resolves to private IPs.
Compatible with the IPv4 and Dualstack IP address types.
Load balancer IP address type Info
Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address types. Public IPv4 addresses have an additional cost.
IPv4
Includes only IPv4 addresses.
Dualstack
Includes IPv4 and IPv6 addresses.
Dualstack without public IPv4
Includes a public IPv6 address, and private IPv4 and IPv6 addresses. Compatible with internet-facing load balancers only.

Network mapping Info

The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.

VPC | Info

The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to Lambda or on-premises targets, or if using VPC peering. To confirm the VPC for your targets, view target groups

vpc-038d6439ed6224f57 (my_vpc)
10.0.0.0/16

Create VPC

IP pools - new | Info

You can optionally choose to configure an IPAM pool as the preferred source for your load balancers IP addresses. Create or view **Pools** in the Amazon VPC IP Address Manager console

☐ Use IPAM pool for public IPv4 addresses

The IPAM pool you choose will be the preferred source of public IPv4 addresses. If the pool is depleted IPv4 addresses will be assigned by AWS.

Availability Zones and subnets | Info

Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to traffic. The load balancer routes traffic to targets in the selected Availability Zones only.

☑ **ap-south-1a (aps1-az1)**

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-0259ff3c30cc325eea
IPv4 subnet CIDR: 10.0.1.0/24 — public_1a

☑ **ap-south-1b (aps1-az3)**

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-0b4a5981482caf8b6
IPv4 subnet CIDR: 10.0.2.0/24 — public_1b

Listeners and routing Info

A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.

▼ Listener HTTP:80 — Remove

Protocol: HTTP

Port: 80
1-65535

Default action | Info

Forward to: MYTARGET — HTTP
Target type: Instance, IPv4

Create target group

Listener tags - *optional*

Consider adding tags to your listener. Tags enable you to categorize your AWS resources so you can more easily manage them.

Add listener tag

You can add up to 50 more tags.

Network Info

For most applications, you can use multiple Availability Zones and let EC2 Auto Scaling balance your instances across the zones. The default VPC and default subnet: suitable for getting started quickly.

VPC

Choose the VPC that defines the virtual network for your Auto Scaling group.

vpc-038d6439ed6224f57 (my_vpc)
10.0.0.0/16

Create a VPC

Availability Zones and subnets

Define which Availability Zones and subnets your Auto Scaling group can use in the chosen VPC.

Select Availability Zones and subnets

aps1-az3 (ap-south-1b) | subnet-044339eb1ed2d00c5 (private) ✕
10.0.3.0/24

Create a subnet

Availability Zone distribution - *new*

Auto Scaling automatically balances instances across Availability Zones. If launch failures occur in a zone, select a strategy.

◉ Balanced best effort
If launches fail in one Availability Zone, Auto Scaling will attempt to launch in another

○ Balanced only
If launches fail in one Availability Zone, Auto Scaling will continue to attempt to la

Network Info

For most applications, you can use multiple Availability Zones and let EC2 Auto Scaling balance your instances across the zones. The default VPC and default subnets suitable for getting started quickly.

VPC

Choose the VPC that defines the virtual network for your Auto Scaling group.

vpc-038d6439ed6224f57 (my_vpc)
10.0.0.0/16

Create a VPC

Availability Zones and subnets

Define which Availability Zones and subnets your Auto Scaling group can use in the chosen VPC.

Select Availability Zones and subnets

aps1-az3 (ap-south-1b) | subnet-044339eb1ed2d00c5 (private)
10.0.3.0/24

Create a subnet

Availability Zone distribution - *new*

Auto Scaling automatically balances instances across Availability Zones. If launch failures occur in a zone, select a strategy.

- (●) Balanced best effort
  If launches fail in one Availability Zone, Auto Scaling will attempt to launch in another
- ( ) Balanced only
  If launches fail in one Availability Zone, Auto Scaling will continue to attempt to la

Auto Scaling groups (1) Info

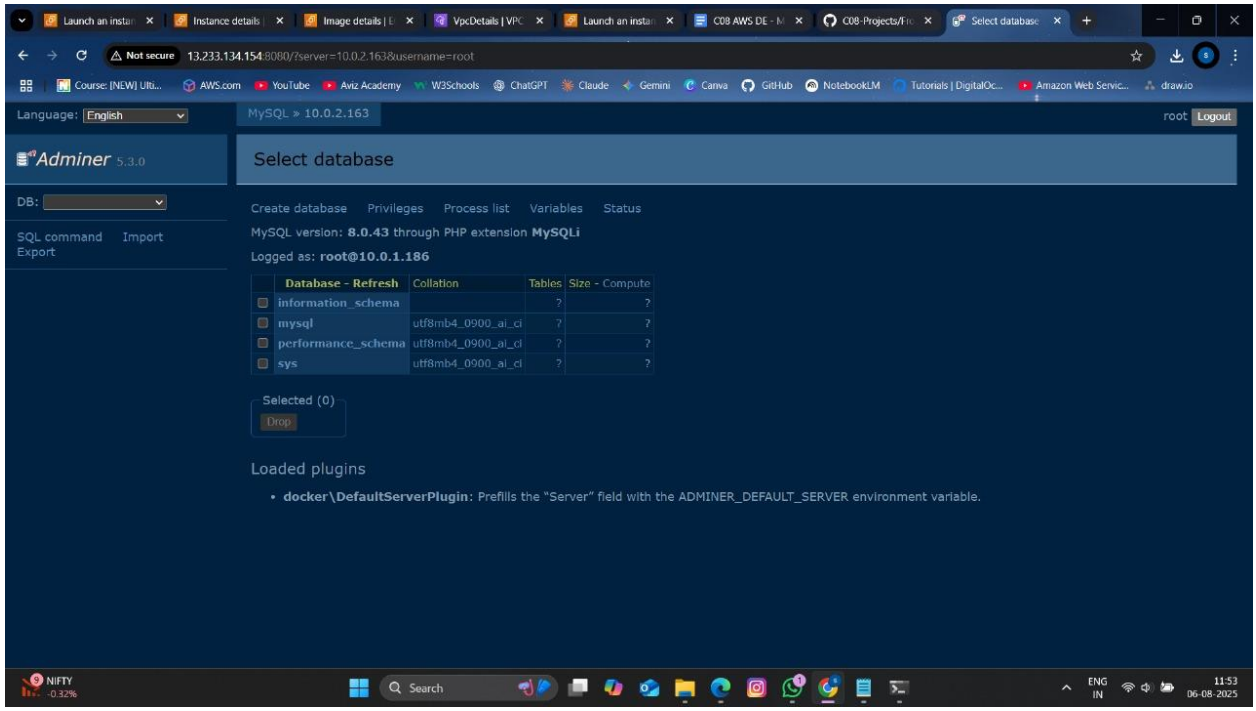
Last updated less than a minute ago | Launch configurations | Launch templates | Actions | Create Auto Scaling group

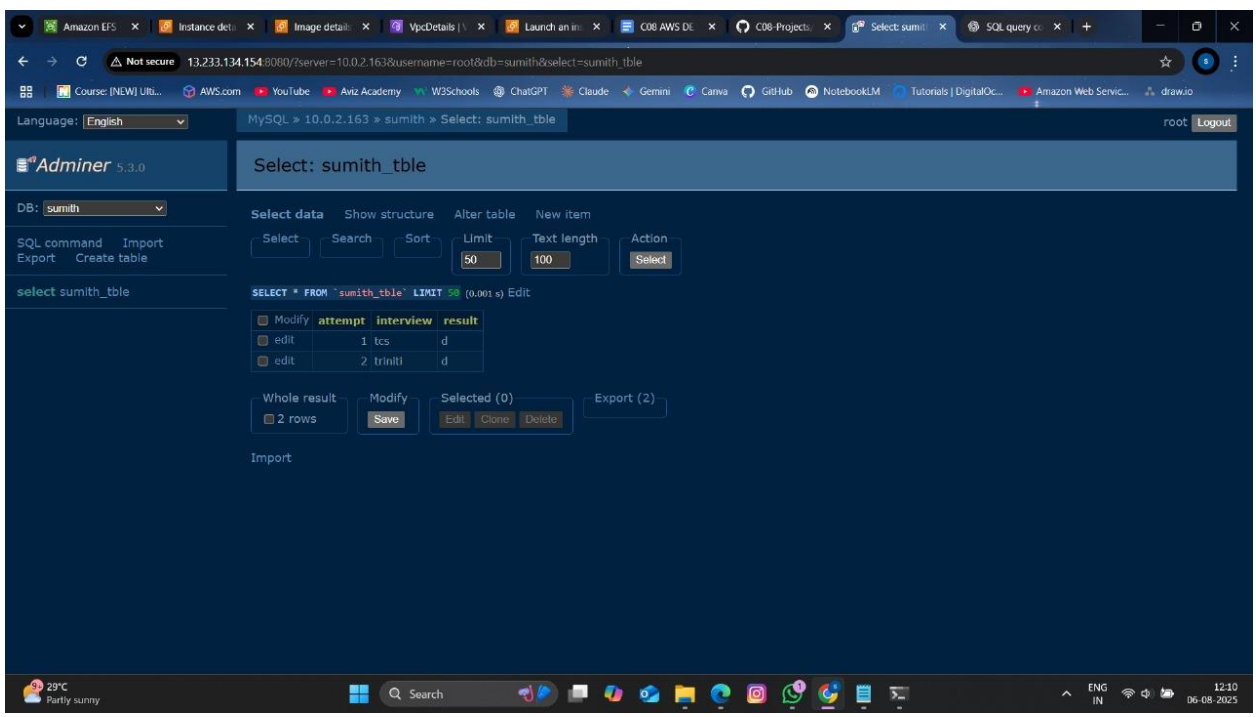
Search your Auto Scaling groups

Name	Launch template/configuration	Instances	Status	Desired capacity	Min	Max	Availability Zones
my_asg	my_temp | Version Default	0	Updating capacity...	1	1	1	aps1-az3 (ap-sout...

C. Validation & Testing

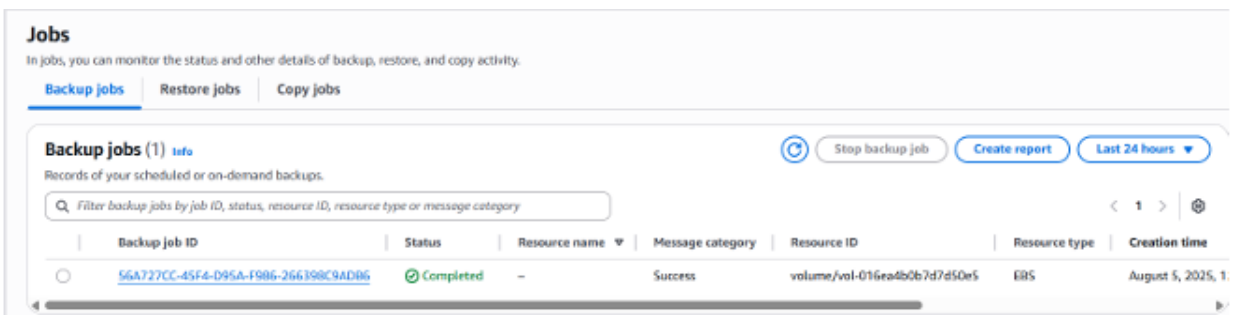
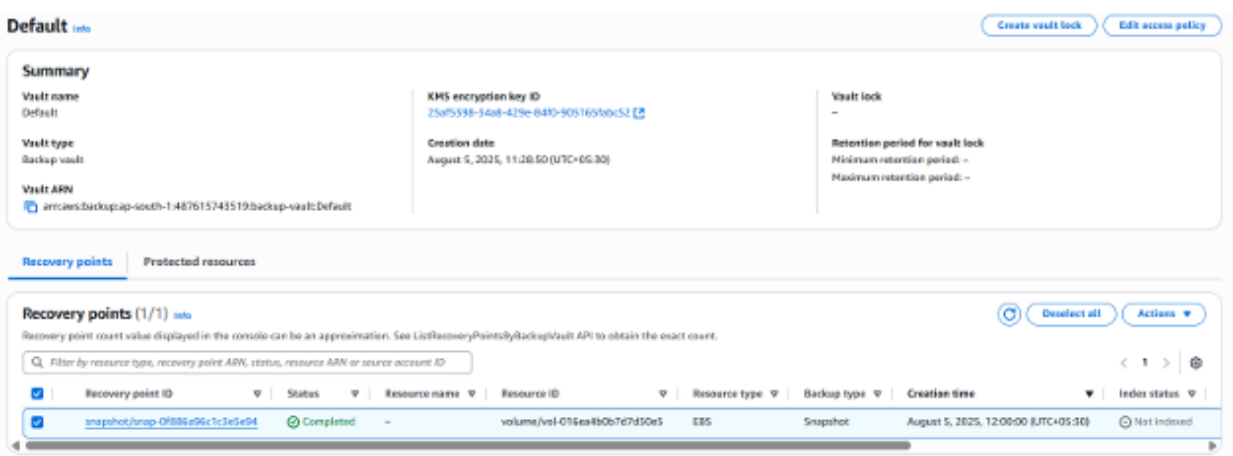
1. **Database Test**: Insert sample data into MySQL and verify persistence after instance restart.





EFS Test: Create a file on one EC2 instance and check accessibility from another.

```
ubuntu@ip-10-0-1-25:~$ sudo touch /mnt/efs/triniti.txt
ubuntu@ip-10-0-1-25:~$ sudo nano /mnt/efs/triniti.txt
ubuntu@ip-10-0-1-25:~$ cat /mnt/efs/triniti.txt
triniti got expelled me.
ubuntu@ip-10-0-1-25:~$ ls /mnt/efs
tcs.txt  triniti.txt
ubuntu@ip-10-0-1-25:~$ sudo cat /mnt/efs/tcs.txt
From Tcs
ubuntu@ip-10-0-1-25:~$
```

```
fs-06437d416070c200b.efs.ap-south-1.amazonaws.com:/  8.0E     0  8.0E   0% /m
ubuntu@ip-10-0-1-129:~$ ls /mnt/efs
triniti.txt
ubuntu@ip-10-0-1-129:~$ sudo cat /mnt/efs/triniti.txt
triniti got expelled me.
ubuntu@ip-10-0-1-129:~$ sudo touch /mnt/efs/tcs.txt
ubuntu@ip-10-0-1-129:~$ echo "From Tcs" | sudo tee /mnt/efs/tcs.txt
From Tcs
ubuntu@ip-10-0-1-129:~$
```

Backup Test: Trigger a manual backup and verify S3 object creation.